Albert Alain Valdivia García

Automation Engineer · Data Engineering & Analytics

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Summary

Automation Engineer with a software background, holding Claude Academy certifications and transitioning into data engineering and analytics. I have built several personal projects that demonstrate my skills in Python and data analysis.

Independent projects

FIFA World Cups 1930–2026

Oct 2026

Web app covering every FIFA World Cup: 23 tournaments, 1,068 matches, 3,028 goals and 84 national teams. · Source on GitHub

- Built with Python and Flask, with a page for every tournament, match and team, plus head-to-head comparisons, SVG charts and a quiz.
- Processed and validated historical data from an external database (CC BY-SA 4.0), checking key figures with tests before importing them.
- Set up automated tests with pytest and Playwright, including accessibility checks (axe-core), run by GitHub Actions on every pull request.
- Configured continuous deployment to Render, with a post-deploy check that the live version is the correct one.
- Made the site trilingual (Spanish, Portuguese and English), with light and dark themes, a mobile-friendly layout and A+ security headers (Mozilla Observatory).
- Followed a professional Git workflow: 69 iterations, each with its own branch, pull request and tests.

Automated Weekly Sales Report

Oct 2026

Python tool that cleans sales files (CSV/Excel), builds an HTML report with charts and emails it every Monday, flagging sharp changes.

- Tested on 540,000 real transactions (UCI Online Retail): processes the full dataset in ~25 seconds and logs every discarded row with its reason.
- Found and fixed errors that skewed the results: shipping fees showed up as the best-selling product (£206,000), cancellations were not subtracted, and an incomplete month looked like a 70% drop.
- Cut false alerts from 8 to 3 over 11 weeks by comparing against a 4-week average instead of the previous week.
- 40 automated tests (pytest) on GitHub Actions; scheduled with Windows Task Scheduler + WSL.

Clinic App: Clinic Management & Booking Marketplace

2026

- Full web application for a multi-branch clinic: 24/7 online booking, scheduling, patient records, point of sale, marketing, inventory, reports and payroll.
- Integrated Stripe payments, SMTP email and Twilio SMS, with a mock mode for development.
- Business logic kept separate from the UI and reused across the app, 5 Jupyter analysis notebooks (pandas, Plotly) and 68 automated tests (pytest).
- Security practices: hashed passwords, lockout after failed login attempts, secrets kept out of the code, and a pilot mode designed with HIPAA healthcare regulations in mind.
- Adapted with real data from a clinic in Florida. Ready to deploy on Render and Streamlit Cloud.
- Built with AI assistance (Claude Code).

Education

B.Eng. Universidad Central "Marta Abreu" de Las Villas, Automation Engineering — Santa Clara, Cuba
2013 – 2017

Skills

Programming & query languages: Python, SQL, Bash

Analysis & visualization: pandas, NumPy, Plotly, Matplotlib, Jupyter

Databases & data warehouses: PostgreSQL, MySQL, SQLite, DuckDB, BigQuery, MongoDB

Web development: Flask, Streamlit, API integrations (Stripe, Twilio, SMTP)

Testing & CI/CD: pytest, Playwright, axe-core, GitHub Actions, continuous deployment on Render and Streamlit Cloud

Applied AI: AI-assisted development with Claude Code

Version control: Git, GitHub (branches and pull requests)

Languages

Spanish: Native

Portuguese: Fluent

English: Intermediate